

PPMI vs OPDC Discovery: Cohort Comparison and Harmonisation Strategy

Cross-cohort replication of ML-derived Parkinson's disease subtypes

Version 3 — 11 March 2026 — All values verified against actual datasets

Abstract

This document compares the PPMI and OPDC Discovery cohorts for cross-cohort replication of data-driven Parkinson's disease subtypes. All sample sizes, clinical scores, and variable availability are verified against the actual datasets used in our analysis (Fichier_PrincipalV61.xlsx for PPMI; OPDC Discovery CSV files for OPDC). Literature-derived values from Marek et al. (2018) and Lawton et al. (2015) are cited separately where relevant for context.

1. Context and Objective

Our initial analysis demonstrated bidirectional replication of a bimodal ($k=2$) severity partition across PPMI and OPDC using five shared features (MDS-UPDRS I, II, III, MoCA, Schwab & England) with CORAL harmonisation and GAMLSS normalisation. Following feedback from Professor Hu, the next phase aims to: (i) expand the shared feature set to include RBD screening questionnaire, olfaction, and orthostatic blood pressure; (ii) test whether $k=3-5$ subtypes can be reliably identified and replicated; and (iii) validate subtypes on imaging biomarkers (SWI, neuromelanin-MRI, DaTscan).

2. Cohort Descriptions (Verified from Data)

Characteristic	PPMI (our dataset)	OPDC Discovery (our dataset)
Total PD in file	821	998 (baseline); 996 (visit 1, used in analysis)
PD used in analysis	794 (after filtering for 5 non-null features)	996 (visit 1 with longitudinal data)
Prodromal / RBD	592 Prodromal	302 polysomnography-confirmed RBD
Healthy Controls	Not in current dataset (196 per literature)	319 (baseline); 316 (visit 1)
Study type	Longitudinal, multicentre (24 sites)	Longitudinal, regional (11 hospitals, Thames Valley)
Diagnostic criteria	Clinical + mandatory DAT-SPECT	UK PD Brain Bank (clinical probability $\geq 90\%$)
Disease stage at entry	De novo, ≤ 2 yrs since diagnosis	Early PD, ≤ 3.5 yrs since diagnosis
Medication status	100% drug-naïve	~85% treated (LEDD>0), ~15% untreated
Follow-up	Up to V21 (~10+ years)	Up to visit 10

3. Baseline Demographics (from actual data)

Note: PPMI values are computed from the 821 PD patients in Fichier_PrincipalV61.xlsx (BL sheet). OPDC values are from 996 PD patients at visit 1.

Variable	PPMI PD (n=821)	OPDC PD (n=996)	Source
Age (years)	Mean 60.7, SD 10.0 (at diagnosis, range 29–85)	Mean 67.4, SD 9.6 (at visit 1, range 32–90)	Computed from data
Age at diagnosis	Mean 60.7 (from PDDXDT-BIRTHDT)	Mean 66.2, SD 9.6 (range 32–90)	Computed
Sex (% male)	59.1% (485/821) [SEX=1 coding]	64.4% (641/996)	Computed
Disease duration (from diagnosis)	Not in General sheet; ~6.7 months per literature	Mean 1.24 yrs, SD 0.94	Computed / Marek 2018
Disease duration (from onset)	Not available	Mean 2.89 yrs, SD 1.85	Computed
Education	Mean 15.7 yrs, SD 3.6	School leaving age: mean 16.4; 61% >12 yrs education	Computed
Untreated patients	821 (100%) — all drug-naïve	146 (14.7%, LEDD=0)	Computed

Variable	PPMI PD (n=821)	OPDC PD (n=996)	Source
Mean LEDD (mg/day)	0	Mean 278.2, SD 209.1	Computed

4. Baseline Clinical Scores (from actual data)

Note: PPMI MDS-UPDRS III is assessed OFF medication (all drug-naïve). OPDC MDS-UPDRS III is assessed in the ON state for treated patients. Schwab & England is available in both cohorts.

Assessment	PPMI (BL sheet, PD)	OPDC PD (visit 1)	n PPMI	n OPDC
MDS-UPDRS I	Available (NP1RTOT)	Mean 8.89, SD 5.27	794	977
MDS-UPDRS II	Available (NP2PTOT)	Mean 8.86, SD 6.10	792	983
MDS-UPDRS III	Available (NP3TOT) (OFF state)	Mean 26.73, SD 10.96 (ON state)	727	979
MDS-UPDRS IV	Available (NP4TOT)	Mean 0.29, SD 1.12	—	984
MoCA	Available (MCATOT)	Mean 24.54, SD 3.52 (adj: 24.91, SD 3.42)	792	985
Schwab & England	Available (MSEADLG)	Mean 88.51, SD 10.04	794	994
Hoehn & Yahr	Available (NHY_OFF)	Mean 1.85, SD 0.55	—	991
ESS (sleepiness)	Not in current file	Mean 7.57, SD 4.49	—	978
RBD Questionnaire	RBDSQ (available in PPMI portal)	Mean 4.86, SD 3.17	To download	947
Olfaction	UPSIT 40 items (available in PPMI portal)	Sniffin' Sticks 16 items; mean 7.03, SD 2.92	To download	951
SBP postural drop	SCOPA-AUT cardiovascular items (PPMI portal)	Mean 6.63, SD 15.99 mmHg	To download	992
DBP postural drop	As above	Mean -2.41, SD 8.97 mmHg	As above	992
HADS Anxiety	STAI (different instrument)	Mean 4.58, SD 3.75	—	978
HADS Depression	GDS-15 (different instrument)	Mean 4.40, SD 3.35	—	972
BDI	Not in dataset	Mean 8.87, SD 6.41	—	933
Purdue Pegboard	Not in dataset	Mean 28.49, SD 6.88	—	984

5. Shared Features for Expanded Clustering

The table below shows verified availability in both cohorts. The first five features were used in the initial analysis; the last three are proposed additions.

Feature	PPMI	OPDC	Harmonisable?	Notes
MDS-UPDRS I	NP1RTOT ✓	UPDRS_I ✓ (n=977)	Yes	Self-reported; same scale
MDS-UPDRS II	NP2PTOT ✓	UPDRS_II ✓ (n=983)	Yes	Self-reported; same scale
MDS-UPDRS III	NP3TOT ✓ (OFF)	UPDRS_III ✓ (n=979, ON)	With caveats	ON vs OFF; restrict to untreated or use GAMLSS correction
MoCA	MCATOT ✓	MOCA_total ✓ (n=985)	Yes	OPDC has education-adjusted version
Schwab & England	MSEADLG ✓	Schwab_england_scale_ ✓ (n=994)	Yes	Same instrument; directly comparable
RBD Questionnaire (NEW)	RBDSQ ✓ (to download from portal)	RBD_total ✓ (n=947)	Yes	Identical RBDSQ instrument; 13-item yes/no
Olfaction (NEW)	UPSIT 40 items ✓ (to download from portal)	Sniffin' Sticks 16 items ✓ (n=951)	Via z-scores	Different tests; harmonise via z-score or percentile rank
Orthostatic BP (NEW)	SCOPA-AUT cardiovascular items ✓ (to download)	SBP/DBP_postural_drop ✓ (n=992)	Partial	PPMI: autonomic questionnaire; OPDC: direct BP measurement

6. Critical Differences for Harmonisation

Medication status: PPMI 100% drug-naïve; OPDC ~85% treated (mean LEDD 278 mg/day). Directly affects UPDRS III comparability. Strategy: restrict to 146 untreated OPDC patients, or apply GAMLSS + CORAL as in initial analysis.

Disease duration: PPMI de novo (≤ 2 yrs); OPDC mean 1.24 yrs from diagnosis (range 0–3.5). Relatively similar windows.

Age: PPMI mean age at diagnosis 60.7 yrs; OPDC mean age at visit 1 is 67.4 yrs (~6 yrs older). GAMLSS normalisation accounts for this.

Diagnostic confirmation: PPMI requires DAT-SPECT (excludes SWEDD); OPDC uses UK PD Brain Bank criteria only. Potential inclusion of non-degenerative parkinsonism in OPDC.

ON/OFF assessment: OPDC has UPDRS-III OFF scores for only 302 subjects at visit 2 (extras file), not routinely collected.

Olfaction scales differ: UPSIT (40-item, forced choice) vs Sniffin' Sticks (16-item, identification). Z-score or percentile transformation required.

7. OPDC Sub-Population Extraction Strategy

7.1 Principle

Identify within OPDC a sub-sample whose clinical profile best approximates PPMI, minimising confounding bias.

7.2 Priority Matching Variables

Disease duration: Restrict OPDC to ≤ 2 yrs from diagnosis.

Medication status: Prioritise 146 untreated patients (LEDD=0). Their UPDRS III is directly comparable to PPMI.

Age at onset: Match to PPMI distribution (mean 60.7, range 29–85).

Sex: Broadly comparable (PPMI 59% male; OPDC 64% male).

7.3 Matching Methods

Method	Principle	Advantage	Limitation
PSM	Logistic model: PPMI vs OPDC, 1:1 matching	Handles multiple covariates	Sample loss; model dependence
CEM	Strata (age ± 5 yrs; duration ± 1 yr) + exact	Simple; no misspecification	Smaller sample possible
Restriction + IPW	Filter OPDC on PPMI criteria, then reweight	Larger sample preserved	Possible extreme weights

7.4 Post-Matching Verification

- SMD < 0.1 for all matched variables.
- QQ plots and density overlays.
- Kolmogorov–Smirnov tests.
- Verify that known biological relationships are preserved.

7.5 Estimated Sample Size

Restricting OPDC to untreated (n=146) with disease duration ≤ 2 yrs yields ~80–120 patients. Including treated patients with LEDD adjustment expands to ~300–400. The choice depends on the priority accorded to direct motor score comparability versus statistical power for higher-k clustering.

8. OPDC Longitudinal Coverage

Visit	PD n	HC n	RBD n
1	996	316	302
2	822	—	—
3	672	—	—
4	515	—	—
5	378	—	—
6	296	—	—
7	175	—	—
8	119	—	—
9	63	—	—
10	24	—	—

Note: HC and RBD longitudinal sample sizes not yet computed per visit from the provided files; available on request.

9. Imaging Data

The following imaging modalities are NOT present in the OPDC clinical CSV files. According to the OPDC-MRI cohort profile (Griffanti et al., 2020, *BMJ Open*), 290 participants have undergone MRI (119 PD, 68 HC, 87 at-risk including RBD). Imaging-derived variables may reside in a separate database.

Modality	PPMI (in our file)	OPDC Clinical CSV	OPDC-MRI Substudy (Griffanti 2020, n=290)
DaTscan (quantitative SUR)	Yes: caudate/putamen R/L (all PD patients)	Not present	Binary result (aj18dat) in clinical data; no quantitative SUR
Grey matter volume	Yes: GM_VOLUME	Not present	T1-weighted structural MRI available; volumes derivable
Subcortical volumes	Yes: caudate, putamen, SN, RN, GPi, striatum	Not present	T1 + FIRST segmentation available
SWI / DNH (nigrosome-1)	Not available	Not present	Available: DNH binary score + quantitative nigrosome-1 signal (~115 subjects)
Neuromelanin-sensitive MRI	Not available	Not present	Available: NM-MRI (~106 subjects)
R2* / QSM (iron mapping)	Not available	Not present	R2* available; QSM derivable from SWI

10. Limitations and Caveats

- PPMI UPDRS III is OFF-state; OPDC is ON-state. Direct comparison requires adjustment or restriction to untreated patients.
- PPMI data file (Fichier_PrincipalV61.xlsx) does not include RBD questionnaire, olfaction, or autonomic measures — these must be downloaded separately from the PPMI portal.

- OPDC has no systematic ON/OFF paired assessments; OFF-state UPDRS III is available for only 302 subjects at visit 2.
- OPDC sex ratio (64% male) is slightly higher than PPMI (59% male); accounted for in matching model.
- OPDC-MRI substudy (n=290) covers only a subset of the clinical cohort; imaging validation will be limited to overlapping subjects.
- Olfaction uses different instruments (UPSIT vs Sniffin' Sticks); z-score harmonisation introduces measurement noise.

References

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